

Study Report: Microsoft Power Platform Developer - Career Path & Core Skills

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Family:	Microsoft Power Platform Developer
Level:	Beginner to Advanced
Tracks:	Beginner, Intermediate, Advanced
Chapters:	18
Source:	Official docs, free libraries, and industry-standard references (see sources and books)

Official sources and free libraries

- <https://learn.microsoft.com/power-platform/>
- <https://learn.microsoft.com/power-apps/>
- <https://learn.microsoft.com/power-automate/>
- <https://learn.microsoft.com/power-bi/>
- <https://learn.microsoft.com/power-platform/alm/>
- <https://learn.microsoft.com/power-platform/guidance/adoption/>

Summary

Microsoft Power Platform developer path spanning canvas and model-driven Power Apps, Power Automate cloud and desktop flows, Dataverse ALM, component libraries, custom connectors, Power BI embed, and enterprise governance. Aligned to PL-400, PL-200, and Microsoft Learn Power Platform documentation.

Skill stack by level

Recommended books

1. Beginning Power Apps - Tim Leung

Level: Beginner | Canvas apps, connectors, galleries, and SharePoint integration.

2. Microsoft Power Platform Documentation - Microsoft Learn

Level: Beginner | Official docs for Power Apps, Automate, BI, and Dataverse.

3. Pro Power BI Architecture - Reza Rad & Milo? Radivojevi?

Level: Advanced | Enterprise BI design, deployment, and governance.

4. Power Automate Documentation - Microsoft Learn

Level: Intermediate | Cloud flows, desktop flows, and approval automation patterns.

5. PL-400 Developer Learning Path - Microsoft Learn

Level: Advanced | Official path for Power Platform developer certification.

End-to-end projects

Project 1: Employee Expense Canvas App

Beginner | 2 weeks - Canvas app for expense submission with SharePoint list backend, approval gallery, and mobile layout.

- Data: SharePoint list created
- App: App screenshot
- Logic: Working submit flow
- Mobile: Mobile screenshot

Project 2: Approval Workflow with Power Automate

Intermediate | 2-3 weeks - Purchase request app + automated approval flow with Teams adaptive cards and Dataverse audit log.

- App: Canvas app
- Flow: Flow diagram export
- Teams: Teams notification screenshot
- Audit: Audit report view

Project 3: CodeQuest Admin Power Platform Suite

Advanced | 4-5 weeks - Model-driven app + embedded Power BI + ALM across environments for school admin reporting.

- Model App: Model app published
- BI Embed: Embedded BI screenshot
- ALM: ALM documentation
- Security: Security test results

Chapters

Track: Beginner

Chapter 1: Power Platform Components Overview [Beginner]

Chapter focus: Power Platform Components Overview** *(Level: Beginner)

Power Platform bundles Power Apps (canvas + model-driven), Power Automate, Power BI, Power Pages, and Copilot Studio on Dataverse and connectors. Makers build low-code solutions; developers extend with code. Licensing varies by per-user and per-app plans.

Code Reference:

```
Get-Command -Module Microsoft.PowerApps.*  
# Explore available admin and maker modules
```

What it shows: PowerShell module discovery supports tenant administration tasks.

Topic guide

What it actually is

Power Platform is Microsoft's low-code suite for business applications.

When to use it

Replacing spreadsheets, automating workflows, and rapid app delivery.

Where to use it

Enterprises on Microsoft 365 and Azure.

Real use example

CodeQuest IT replaces Excel roster with SharePoint + canvas app in two sprints.

Chapter 2: Canvas App Controls and Formulas [Beginner]

Chapter focus: Canvas App Controls and Formulas** *(Level: Beginner)

Canvas apps use Excel-like formulas: Filter, LookUp, Patch, Collect, Gallery controls, forms, and navigation. Delegation warnings mean formulas may not run server-side-understand data source limits for SharePoint and Dataverse.

Code Reference:

```
// Filter(Devices, Status = "Needs Repair")  
// Patch(Inspections, Defaults(Inspections), {DeviceId: galDevices.Selected.ID})
```

What it shows: Filter and Patch patterns are daily canvas app bread and butter.

Topic guide

What it actually is

Canvas apps offer flexible pixel-perfect mobile and web UX.

When to use it

Field apps, approvals, and lightweight replacements for paper forms.

Where to use it

Frontline workers, events, and inventory walks.

Real use example

Teacher inspection app captures photo and notes per classroom device.

Chapter 3: Data Sources: SharePoint and Excel [Beginner]

Chapter focus: Data Sources: SharePoint and Excel** *(Level: Beginner)

Makers often start with SharePoint lists and Excel tables in OneDrive. Know column types, choice columns, and permissions inheritance. Migrate to Dataverse when needing relational model, security roles, or ALM.

Code Reference:

```
# SharePoint list: Inspections | Columns: Title, DeviceId, Photo, Score, InspectorEmail
```

What it shows: List schema comments guide citizen developer setup.

Topic guide

What it actually is

SharePoint is the quickest citizen-developer data store.

When to use it

Team-level apps before enterprise ALM investment.

Where to use it

Departments already on Microsoft 365.

Real use example

HR attendance list backs canvas app until row count triggers Dataverse migration.

Chapter 4: Power Automate Flow Triggers and Actions [Beginner]

Chapter focus: Power Automate Flow Triggers and Actions** *(Level: Beginner)

Flows start with triggers: manual button, scheduled recurrence, SharePoint item created, Dataverse row updated, or HTTP webhook. Actions chain connectors with dynamic content tokens. Use Compose and variables for intermediate values.

Code Reference:

```
# Trigger: When an item is created (SharePoint)
# Action: Send email + Create row (Dataverse)
```

What it shows: Trigger-action outline documents common flow skeleton.

Topic guide

What it actually is

Power Automate connects hundreds of SaaS and on-prem systems.

When to use it

Email alerts, approvals, sync, and scheduled maintenance.

Where to use it

Every Power Platform tenant.

Real use example

New inspection item triggers manager approval before repair ticket creation.

Chapter 5: Introduction to Dataverse [Beginner]

Chapter focus: Introduction to Dataverse** *(Level: Beginner)

Dataverse provides relational tables, business rules, plugins, and enterprise security. Premium connectors and per-app licensing often require Dataverse. Use solutions for metadata packaging from day one on team projects.

Code Reference:

```
pac org who # displays current Dataverse environment and user
```

What it shows: PAC CLI confirms target environment before solution operations.

Topic guide

What it actually is

Dataverse is the enterprise data backbone of Power Platform.

When to use it

Apps needing role-based security, complex relationships, or ALM.

Where to use it

Dynamics apps and standalone Power Apps.

Real use example

Team graduates canvas prototype to Dataverse with proper cq_ publisher prefix.

Chapter 6: Power BI Basics for App Makers [Beginner]

Chapter focus: Power BI Basics for App Makers** *(Level: Beginner)

Power BI Desktop imports or connects to data, models relationships, and publishes reports to service. App makers embed tiles or reports in Teams and model-driven apps. Understand import vs DirectQuery vs Dataverse connector trade-offs.

Code Reference:

```
# Measure: Completion Rate = DIVIDE(COUNTROWS(FILTER(Enrollments, Status="Complete")),  
COUNTROWS(Enrollments))
```

What it shows: DAX measure example calculates course completion KPI.

Topic guide

What it actually is

Power BI visualizes data for decisions inside Microsoft ecosystem.

When to use it

Executive dashboards complement operational canvas apps.

Where to use it

Organizations standardized on Microsoft analytics.

Real use example

Principal views embedded completion report inside Teams channel.

Track: Intermediate

Chapter 7: Model-Driven Apps vs Canvas Apps [Intermediate]

Chapter focus: Model-Driven Apps vs Canvas Apps** *(Level: Intermediate)

Choose model-driven when process-centric with complex Dataverse schema; canvas for tailored UX and mixed connectors. Hybrid patterns embed canvas pages in model-driven apps. Align choice to maintainability and team skills.

Code Reference:

```
# Decision: Dataverse-heavy CRM process -> Model-driven
# Field inspection UX -> Canvas
```

What it shows: Decision comments prevent wrong app type selection.

Topic guide

What it actually is

App type selection affects delivery speed and long-term maintenance.

When to use it

Project kickoff architecture decisions.

Where to use it

Consulting engagements and internal COE reviews.

Real use example

COE steers district portal to model-driven; warehouse scanner stays canvas.

Chapter 8: Component Libraries and Reusable UX [Intermediate]

Chapter focus: Component Libraries and Reusable UX** *(Level: Intermediate)

Component libraries share screens and controls across apps. Publish updates propagate to dependent apps-plan versioning. Use input/output properties for parameterized components.

Code Reference:

```
// Component: HeaderBar | Inputs: TitleText, UserName | Outputs: OnHomeSelect
```

What it shows: Component contract documents reusable UI modules.

Topic guide

What it actually is

Component libraries enforce consistent UX and reduce duplication.

When to use it

Multi-app portfolios in large tenants.

Where to use it

Enterprises with branded app standards.

Real use example

HeaderBar component update rolls branding fix to twelve apps automatically.

Chapter 9: Power Automate Advanced Patterns [Intermediate]**Chapter focus: Power Automate Advanced Patterns** *(Level: Intermediate)**

Child flows promote reuse. Try-Catch scopes handle errors. HTTP with Azure AD OAuth calls custom APIs. Desktop flows automate legacy Win32 apps alongside cloud flows in attended/unattended mode.

Code Reference:

```
# Child flow: Resolve-ManagerEmail | Input: UserId | Output: ManagerEmail
```

What it shows: Child flow signatures document reusable automation modules.

Topic guide**What it actually is**

Advanced flow patterns scale automation reliably.

When to use it

Enterprise integrations and RPA hybrids.

Where to use it

Finance, HR, and IT operations automation.

Real use example

Desktop flow reads legacy grading system export; cloud flow imports to Dataverse.

Chapter 10: Custom Connectors and Azure API Management [Intermediate]**Chapter focus: Custom Connectors and Azure API Management** *(Level: Intermediate)**

Custom connectors wrap OpenAPI definitions for proprietary APIs. APIM policies add rate limit, JWT validation, and logging. Store secrets in Azure Key Vault referenced by connector.

Code Reference:

```
# OpenAPI: POST /enrollments { studentId, courseId } -> 201 Created
```

What it shows: OpenAPI snippet defines connector action contract.

Topic guide**What it actually is**

Custom connectors extend Power Platform to any REST API.

When to use it

Proprietary LMS, billing, and internal microservices.

Where to use it

Organizations with API-first backends.

Real use example

CodeQuest enrollment API exposed as custom connector for canvas app Patch calls.

Chapter 11: Dataverse Business Rules and Low-Code Logic [Intermediate]

Chapter focus: Dataverse Business Rules and Low-Code Logic** *(Level: Intermediate)

Business rules show/hide fields, set defaults, and validate without code. Combine with calculated and rollup columns. When rules conflict with plugins, document execution order.

Code Reference:

```
# Rule: If Status = Inactive -> Lock Email field | Scope: Entity + Form
```

What it shows: Business rule scope comments clarify where logic applies.

Topic guide

What it actually is

Business rules keep simple logic maintainable by makers.

When to use it

Field validation and dynamic forms on Dataverse.

Where to use it

Functional consultants configuring CRM apps.

Real use example

Inactive contact rule prevents accidental outreach during audit hold.

Chapter 12: ALM: Solutions, Layers, and Pipelines [Intermediate]

Chapter focus: ALM: Solutions, Layers, and Pipelines** *(Level: Intermediate)

Split solutions: base (managed), patch, and optional unmanaged dev layer. Environment variables hold config; connection references abstract auth. Pipelines deploy with approval gates and automated tests.

Code Reference:

```
pac solution import --path ./CodeQuestCore_managed.zip --environment https://test.crm.dynamics.com
```

What it shows: PAC import command is CI/CD building block for releases.

Topic guide

What it actually is

ALM disciplines prevent production-breaking citizen changes.

When to use it

Multi-environment enterprise programs.

Where to use it

Regulated industries requiring change audit trails.

Real use example

Pipeline rejects import when Solution Checker finds critical security issue.

Chapter 13: Monitoring with Power Platform Admin Center [Intermediate]

Chapter focus: Monitoring with Power Platform Admin Center *(Level: Intermediate)**

Admin center shows environment health, capacity, DLP policies, and analytics. Monitor flow run failures, connector throttling, and premium request consumption. Set alerts for unused apps and orphaned connections.

Code Reference:

```
Get-AdminFlow | Where-Object {$_.Enabled -eq $true -and $_.State -eq 'Suspended'}
```

What it shows: PowerShell helps audit flows needing owner reassignment.

Topic guide

What it actually is

Tenant monitoring prevents outages and security drift.

When to use it

COE ongoing operations.

Where to use it

Large tenants with hundreds of makers.

Real use example

Weekly report flags flows owned by departed users for cleanup.

Track: Advanced

Chapter 14: Power Apps Component Framework (PCF) [Advanced]

Chapter focus: Power Apps Component Framework (PCF) *(Level: Advanced)**

PCF builds custom controls with TypeScript for model-driven and canvas apps. Use for specialized visuals, masked inputs, or integrating third-party JS libraries not available natively. Bundle with solution and document manifest features.

Code Reference:

```
// PCF manifest: bound property Dataset | React virtualized grid for 10k rows
```

What it shows: PCF manifest comment describes control binding strategy.

Topic guide

What it actually is

PCF extends UI when native controls insufficient.

When to use it

Complex grids, maps, and bespoke visualizations.

Where to use it

Pro-dev teams supporting citizen makers.

Real use example

Virtualized enrollment grid PCF replaces slow default subgrid on large districts.

Chapter 15: Power BI Embed and Row-Level Security [Advanced]

Chapter focus: Power BI Embed and Row-Level Security** *(Level: Advanced)

Embed reports in apps using user-owns-data or app-owns-data models. Map Dataverse teams to Power BI RLS roles with USERNAME() or custom DAX. Manage embed token refresh and capacity SKUs for production SLAs.

Code Reference:

```
// RLS: [Email] = USERNAME() for manager-only district view
```

What it shows: RLS DAX ties Power BI rows to logged-in maker identity.

Topic guide

What it actually is

Embedded analytics deliver contextual insights inside apps.

When to use it

Executive and manager dashboards in CRM apps.

Where to use it

SaaS products white-labeling analytics for customers.

Real use example

District manager sees only their schools in embedded completion dashboard.

Chapter 16: Environment Variables and Secret Management [Advanced]

Chapter focus: Environment Variables and Secret Management** *(Level: Advanced)

Environment variables store text, JSON, and secret values per environment. Reference in flows and plugins instead of hardcoding. Rotate secrets with pipeline automation and document in

runbooks-never commit secrets to Git.

Code Reference:

```
# Variable:    cq_ApiBaseUrl    (text)    |    cq_ApiKey    (secret)    |    current:
https://api.test.codequest.app
```

What it shows: Variable inventory documents ALM-safe configuration.

Topic guide

What it actually is

Environment variables enable portable solutions across stages.

When to use it

Multi-stage ALM and ISV packaged solutions.

Where to use it

Partners shipping managed solutions to customers.

Real use example

Prod secret rotated quarterly via pipeline without editing flows manually.

Chapter 17: Performance and Delegation Strategy [Advanced]

Chapter focus: Performance and Delegation Strategy** *(Level: Advanced)

Canvas delegation limits Filter/Search on large Dataverse sets-pre-filter with views or use StartsWith on delegable columns. Monitor Premium API usage. Archive historical data to keep apps fast.

Code Reference:

```
// Delegable: Filter(Accounts, StartsWith(Name, "North"))
// Non-delegable warning: Search() on 50k rows
```

What it shows: Delegable vs non-delegable examples teach formula fixes.

Topic guide

What it actually is

Performance work keeps apps usable at enterprise data volumes.

When to use it

Slow galleries and timeout errors in production apps.

Where to use it

Apps exceeding 2000 row delegation threshold.

Real use example

View pre-filtered to active districts fixes gallery load timeout.

Chapter 18: Copilot Studio and AI Builder in Solutions [Advanced]

Chapter focus: Copilot Studio and AI Builder in Solutions** *(Level: Advanced)

Copilot Studio agents orchestrate topics with Dataverse and connector actions. AI Builder provides prebuilt models for prediction and form processing. Govern AI with DLP, logging, and human escalation paths for wrong answers.

Code Reference:

```
# Agent topic: ResetPassword -> Verify identity -> Power Automate AD reset flow
```

What it shows: Topic flow ties conversational AI to governed automation.

Topic guide

What it actually is

AI services extend Power Platform with NLP and document intelligence.

When to use it

Self-service portals and document-heavy workflows.

Where to use it

Enterprises adopting Microsoft Copilot stack.

Real use example

AI Builder extracts invoice fields into Dataverse; agent handles employee FAQs.